

**INSTITUTE OF ENGINEERING AND TECHNOLOGY
DEEN DAYAL UPADHYAYA GORAKHPUR UNIVERSITY
GORAKHPUR**



**COURSE STRUCTURE
&
SYLLABUS**

FOR

B.TECH.

in

Artificial Intelligence and Machine Learning

Based

on

AICTE MODEL CURRICULUM

[Effective from the Session: 2025-26]

Semester-VI (Third Year) Curriculum

S. No.	Type of Course	Course Code	Course Title	Credit
1.	Professional Core Course	AIML306	Distributed System	3+0
2.	Professional Core Course	AIML307	Neural Network & Deep Learning	3+0
3.	Professional Core Course	AIML308	Cloud Computing	3+0
4.	Professional Core Course	AIML309	Image Processing	3+0
5.	Professional Elective Course		Elective-II (from SWAYAM/ Department of AIML)	3+0
6.	Open Elective Course		Open Elective-I (from Department) or Open Elective-I (from SWAYAM)	3+0
7.	Humanities & Social Science including Management Courses	HUV102	Universal Human Values-II: Understanding Harmony and Ethical Human Conduct	3+0
8.	Professional Core Course	AIML357	Neural Network & Deep Learning Lab	0+1
9.	Professional Core Course	AIML358	Cloud Computing Lab	0+1
10.	Project work, Seminar and internship in industry or elsewhere	AIMLMP2	AIML Mini Project-II	0+2
Total Credit				25

Distributed System (AIML306)

Credit: 3+0

Course Objective: This course is an introduction to the design of distributed systems and algorithms that support distributed computing.

Syllabus

- 1. Characterization of Distributed Systems:** Introduction, Examples of distributed Systems, Resource sharing and the Web Challenges. Architectural models, Fundamental Models. Theoretical Foundation for Distributed System: Limitation of Distributed system, absence of global clock, shared memory, Logical clocks, Lamport's & vectors logical clocks. Concepts in Message Passing Systems: causal order, total order, total causal order, Techniques for Message Ordering, Causal ordering of messages, global state, and termination detection.
- 2. Distributed Mutual Exclusion:** Classification of distributed mutual exclusion, requirement of mutual exclusion theorem, Token based and non-token-based algorithms, performance metric for distributed mutual exclusion algorithms. Distributed Deadlock Detection: system model, resource Vs communication deadlocks, deadlock prevention, avoidance, detection & resolution, centralized deadlock detection, distributed deadlock detection, path pushing algorithms, edge chasing algorithms.
- 3. Agreement Protocols:** Introduction, System models, classification of Agreement Problem, Byzantine agreement problem, Consensus problem, Interactive consistency Problem, Solution to Byzantine Agreement problem, Application of Agreement problem, Atomic Commit in Distributed Database system. Distributed Resource Management: Issues in distributed File Systems, Mechanism for building distributed file systems, Design issues in Distributed Shared Memory, Algorithm for Implementation of Distributed Shared Memory.
- 4. Failure Recovery in Distributed Systems & Transactions and Concurrency Control:** Concepts in Backward and Forward recovery, Recovery in Concurrent systems, obtaining consistent Checkpoints, Recovery in Distributed Database Systems. Fault Tolerance: Issues in Fault Tolerance, Commit Protocols, Voting protocols, Dynamic voting protocols. Transactions and Concurrency Control: Transactions, Nested transactions, Locks, Optimistic Concurrency control, Timestamp ordering, Comparison of methods for concurrency control. Distributed Transactions: Flat and nested distributed transactions, Atomic Commit protocols, Concurrency control in distributed transactions, Distributed deadlocks, Transaction recovery. Replication: System model and group communication, Fault - tolerant services, highly available services, Transactions with replicated data.

Course Outcome (CO) : At the end of course, the student will be able to

CO1: Provide hardware and software issues in modern distributed systems.

CO2: Get knowledge in distributed architecture, naming, synchronization, consistency and replication, fault tolerance, security, and distributed file systems.

CO3: Analyze the current popular distributed systems such as peer-to-peer (P2P) systems will also be analyzed.

CO4: Know about Shared Memory Techniques and have sufficient knowledge about file access.

References

1. Singhal & Shivaratri, "Advanced Concept in Operating Systems", McGraw Hill
2. Ramakrishna, Gehrke, "Database Management Systems", McGraw Hill
3. Vijay K. Garg Elements of Distributed Computing, Wiley
4. Coulouris, Dollimore, Kindberg, "Distributed System: Concepts and Design", Pearson Education
5. Tenenbaum, Steen, "Distributed Systems", PHI

Neural Network & Deep Learning (AIML307)

Credit: 3+0

Course Objective: Deep learning is a class of machine learning algorithms which enables computers to learn from examples. Deep learning techniques have been used successfully for a variety of applications, including: automatic speech recognition, image recognition, natural language processing, drug discovery, and recommendation systems. In this course, students will learn the fundamentals of deep learning, and the main research activities in this field.

Syllabus

1. **Neural Network:** Neural Network Concepts and Paradigms- What Neural Networks are? Why they are useful, Neural network components and terminology- Topologies - Adaptation, Comparing neural networks and other information Processing methods- Stochastic- Kalman filters - Linear and Nonlinear regression - Correlation - Bayes classification -Vector quantization - Radial basis functions Preprocessing - Post processing.
2. **Introduction to Deep Networks:** Introduction to machine learning- Linear models (SVMs and Perceptrons, logistic regression)- Intro to Neural Nets: What a shallow network computes- Training a network: loss functions, back propagation and stochastic gradient descent- Neural networks as universal function approximates. DEEP NETWORKS: History of Deep Learning- A Probabilistic Theory of Deep Learning- Backpropagation and regularization, batch normalization- VC Dimension and Neural Nets-Deep Vs Shallow Networks-Convolutional Networks- Generative Adversarial Networks (GAN), Semi-supervised Learning.
3. **Dimensionality Reduction:** Linear (PCA, LDA) and manifolds, metric learning – Auto encoders and dimensionality reduction in networks - Introduction to Convnet - Architectures – AlexNet, VGG, Inception, ResNet - Training a Convnet: weights initialization, batch normalization, hyper parameter optimization.
4. **Optimization and Generalization:** Optimization in deep learning– non-convex optimization for deep networks- Stochastic Optimization Generalization in neural networks- Spatial Transformer Networks- Recurrent networks, LSTM - Recurrent Neural Network Language Models- Word-Level RNNs & Deep Reinforcement Learning - Computational & Artificial Neuroscience. Case Study and Applications: Image net- Detection-Audio Wave Net-Natural Language Processing Word2Vec - Joint Detection-Bioinformatics- Face Recognition- Scene Understanding- Gathering Image Captions.

Course Outcome (CO): At the end of course, the student will be able

CO1: To present the mathematical, statistical and computational challenges of building neural networks.

CO2: To study the concepts of deep learning.

CO3: To introduce dimensionality reduction techniques.

CO4: To enable the students to know deep learning techniques to support real-time applications.

References

1. Cosma Rohilla Shalizi, Advanced Data Analysis from an Elementary Point of View, 2015.
2. Deng & Yu, Deep Learning: Methods and Applications, Now Publishers, 2013.
3. Ian Goodfellow, Yoshua Bengio, Aaron Courville, Deep Learning, MIT Press, 2016.
4. Michael Nielsen, Neural Networks and Deep Learning, Determination Press, 2015.

Neural Network & Deep Learning Lab (AIML357)

Credit: 1+0

1. Build a deep neural network model start with linear regression using a single variable.
2. Build a deep neural network model start with linear regression using multiple variables.
3. Write a program to convert speech into text.
4. Write a program to convert text into speech.
5. Write a program for Time-Series Forecasting with the LSTM Model.
6. Implementation of CNN
7. Implementation of RNN
8. Implementation of RESNET
9. Implementation of ALEXNET
10. Implementation of VGG 16.

Cloud Computing (AIML308)

Credit: 3+0

Course Objective: This course provides the students an insight into the basics of cloud computing along with virtualization, cloud computing is one of the fastest growing domains from a while now. It will provide the students basic understanding about cloud and virtualization along with how one can migrate over it.

Syllabus

- 1. Introduction to Cloud Computing:** Definition of Cloud – Evolution of Cloud Computing – Underlying Principles of Parallel and Distributed Computing – Cloud Characteristics – Elasticity in Cloud – On-demand Provisioning.
- 2. Cloud Enabling Technologies Service Oriented Architecture:** REST and Systems of Systems – Web Services – Publish, Subscribe Model – Basics of Virtualization – Types of Virtualizations – Implementation Levels of Virtualization – Virtualization Structures – Tools and Mechanisms – Virtualization of CPU – Memory – I/O Devices – Virtualization Support and Disaster Recovery.
- 3. Cloud Architecture, Services and Storage:** Layered Cloud Architecture Design – NIST Cloud Computing Reference Architecture – Public, Private and Hybrid Clouds – IaaS
- 4. IaaS - PaaS – SaaS – Architectural Design Challenges – Cloud Storage – Storage-as-a-Service – Advantages of Cloud Storage – Cloud Storage Providers – S3.**
- 5. Resource Management and Security in Cloud & Cloud Technologies and Advancements**
Hadoop: Inter Cloud Resource Management – Resource Provisioning and Resource Provisioning Methods – Global Exchange of Cloud Resources – Security Overview – Cloud Security Challenges – Software-as-a-Service Security – Security Governance – Virtual Machine Security – IAM – Security Standards. **Cloud Technologies and Advancements Hadoop:** MapReduce – Virtual Box – Google App Engine – Programming Environment for Google App Engine – Open Stack – Federation in the Cloud – Four Levels of Federation – Federated Services and Applications – Future of Federation.

Course Outcome (CO): At the end of course, the student will be able to

CO1: Describe architecture and underlying principles of cloud computing.

CO2: Explain the need, types and tools of Virtualization for cloud.

CO3: Describe Services Oriented Architecture and various types of cloud services.

CO4: Explain Inter cloud resources management cloud storage services and their providers Assess security services and standards for cloud computing.

References

1. Kai Hwang, Geoffrey C. Fox, Jack G. Dongarra, “Distributed and Cloud Computing, From Parallel Processing to the Internet of Things”, Morgan Kaufmann Publishers, 2012.
2. Rittinghouse, John W., and James F. Ransome, —Cloud Computing: Implementation, Management and Security, CRC Press, 2017.
3. Rajkumar Buyya, Christian Vecchiola, S. ThamaraiSelvi, —Mastering Cloud Computing, Tata Mcgraw Hill, 2013.
4. Toby Velte, Anthony Velte, Robert Elsenpeter, “Cloud Computing – A Practical Approach, Tata Mcgraw Hill, 2009.
5. George Reese, “Cloud Application Architectures: Building Applications and Infrastructure in the Cloud: Transactional Systems for EC2 and Beyond (Theory in Practice), O’Reilly, 2009.

Cloud Computing Lab(AIML358)

Credit: 1+0

1. Install Virtualbox/VMware Workstation with different flavours of linux or windows OS on top of windows7 or 8.
2. Install a C compiler in the virtual machine created using virtual box and execute Simple Programs
3. Install Google App Engine. Create hello world app and other simple web applications using python/java.
4. Use GAE launcher to launch the web applications.
5. Simulate a cloud scenario using CloudSim and run a scheduling algorithm that is not present in CloudSim.
6. Find a procedure to transfer the files from one virtual machine to another virtual machine.
7. Find a procedure to launch virtual machine using trystack (Online Openstack Demo Version)
8. Install Hadoop single node cluster and run simple applications like wordcount.

Image Processing (AIML309)

Credit: 3+0

Course Objective: Students will be able to understand the basic concepts of two-dimensional signal acquisition, sampling, quantization and color model. Apply image processing techniques for image enhancement in both the spatial and frequency domains.

1. DIGITAL IMAGE FUNDAMENTALS: Steps in Digital Image Processing – Components – Elements of Visual Perception – Image Sensing and Acquisition – Image Sampling and Quantization – Relationships between pixels – Color image fundamentals – RGB, HSI models, Two-dimensional mathematical preliminaries, 2D transforms – DFT, DCT.

2. IMAGE ENHANCEMENT: Spatial Domain: Gray level transformations – Histogram processing – Basics of Spatial Filtering– Smoothing and Sharpening Spatial Filtering, Frequency Domain: Introduction to Fourier Transform– Smoothing and Sharpening frequency domain filters – Ideal, Butterworth and Gaussian filters, Homomorphic filtering, Color image enhancement.

3. IMAGE RESTORATION AND IMAGE SEGMENTATION: Image Restoration – degradation model, Properties, Noise models – Mean Filters – Order Statistics – Adaptive filters – Band reject Filters – Band pass Filters – Notch Filters – Optimum Notch Filtering – Inverse Filtering – Wiener filtering
IMAGE SEGMENTATION: Edge detection, Edge linking via Hough transform – Thresholding – Region based segmentation – Region growing – Region splitting and merging – Morphological processing- erosion and dilation, Segmentation by morphological watersheds – basic concepts – Dam construction – Watershed segmentation algorithm.

4. IMAGE COMPRESSION AND RECOGNITION: Need for data compression, Huffman, Run Length Encoding, Shift codes, Arithmetic coding, JPEG standard, MPEG. Boundary representation, Boundary description, Fourier Descriptor, Regional Descriptors – Topological feature, Texture – Patterns and Pattern classes – Recognition based on matching.

References

1. Rafael C. Gonzalez, Richard E. Woods, Digital Image Processing Pearson, 3rd Edition, 2010
2. Anil K. Jain, Fundamentals of Digital Image Processing Pearson, 2002.
3. Kenneth R. Castleman, Digital Image Processing Pearson, 2006.
4. Rafael C. Gonzalez, Richard E. Woods, Steven Eddins, Digital Image Processing using MATLAB Pearson Education, Inc., 2011.